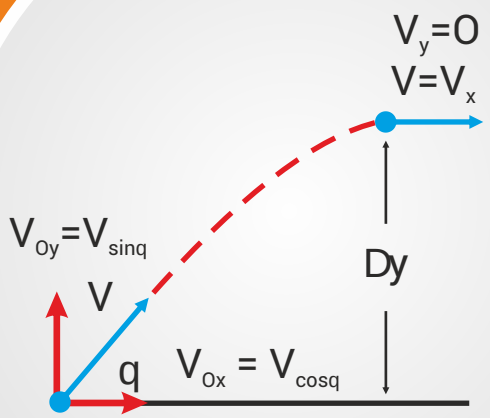
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success

PHYSICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Kinematics-2D

(English Medium)

EXERCISE (O)

SINGLE CORRECT TYPE QUESTIONS

General 2-D motion

1. A body starts with a velocity $(2\hat{i}+3\hat{j}+11\hat{k})$ m/ s and moves with an acceleration $(5\hat{i}+5\hat{j}-5\hat{k})$ m/ s^2. What is its velocity after 0.2 sec?

(A) $7\hat{i}+8\hat{j}+6\hat{k}$ (B) $2\hat{i}-3\hat{j}+11\hat{k}$ (C) $3\hat{i}-4\hat{j}-10\hat{k}$ (D) $3\hat{i}+4\hat{j}+10\hat{k}$

KM0329

2. The position vector of a particle is given by $\vec{r}=(3t^2\hat{i}+4t^2\hat{j}+7\hat{k})$m at a given time t . The net displacement of the particle after 10 s is:

(A) 500m (B) 400m (C) 300m (D) 700m

KM0330

3. A particle starts from the origin at t = 0s with a velocity of $10.0\hat{j}$ m/s and moves in the xy - plane with a constant acceleration of $(8\hat{i}+2\hat{j})$ ms^{-2}. Then y– coordinate of the particle in 2 sec is:

(A) 24 m (B) 16 m (C) 8 m (D) 12 m

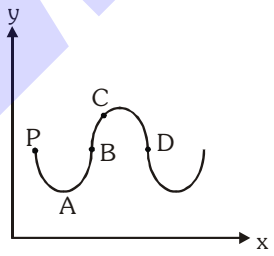
KM0331

4. A car moving at a constant speed of 36 kmph moves north wards for 20 minutes then due to west with the same speed for $8\frac{1}{3}$ minutes. What is the average velocity of the car during this run in kmph.

(A) 27.5 (B) 40.5 (C) 20.8 (D) 32.7

KM0332

5. A man moves in x-y plane along the path shown. At what point is his average velocity vector is in the same direction as his instaneous velocity vector. The man starts from point P.



(A) A (B) B (C) C (D) D

KM0333

6. The position vector of a particle is determined by $\vec{r}=3t^2\hat{i}+4t^2\hat{j}+7\hat{k}$. The distance travelled in first 10 sec is :-

(A) 100 m (B) 150 m (C) 500 m (D) 300 m

KM0186

7. A particle moves in the x-y plane. It x and y coordinates vary with time t according to equations $x = t^2 + 2t$ and $y = 2t$. Possible shape of path followed by the particle is

(A) Straight line (B) Circle
(C) Parabola (D) More information is required to decide.

KM0188

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8. A body of mass 5 kg starts from the origin with an initial velocity $\vec{u} = \left(30\hat{i} + 40\hat{j}\right) ms^{-1}$. If a constant force $\left(-6\hat{i} - 5\hat{j}\right) N$ acts on the body, the time in which the y component of the velocity becomes zero, is :-

(A) 5s (B) 20 s (C) 40 s (D) 80 s

KM0154

Projectile motion

9. The velocity of projection of a projectile is $(6\hat{i} + 8\hat{j})$ ms^{-1}. The horizontal range of the projectile is :

(A) 4.9 m (B) 9.6 m (C) 19.6 m (D) 14 m

KM0334

10. A projectile can have the same range R for two angles of projection. If t_1 and t_2 be the times of flight in two cases, then the product of two times of flight is :

(A) $t_1t_2 \propto R^2$ (B) $t_1t_2 \propto R$ (C) $t_1t_2 \propto \frac{1}{R}$ (D) $t_1t_2 \propto \frac{1}{R^2}$

KM0335

11. A particle is projected from the ground with velocity u at angle θ with horizontal. The horizontal range, maximum height and time of flight are R, H and T respectively. They are given by

$R = \frac{u^2 \sin 2\theta}{g}$ $H = \frac{u^2 \sin^2 \theta}{2g}$ and $T = \frac{2u \sin \theta}{g}$

Now keeping u fixed, θ is varied from 30° to 60°, then

(A) R will first increase then decrease, H will increase and T will decrease

(B) R will first increase then decrease while H and T both will increase

(C) R will decrease while H and T both will increase

(D) R will increase while H and T both will also increase **KM0156**

12. A projectile is given an initial velocity of $(\hat{i} + 2\hat{j})$ m/sec. The cartesian equation of its path is : (g = 10 m/s^2)

(A) $y = 2x - 5x^2$ (B) $y = x - 5x^2$ (C) $4y = 2x - 5x^2$ (D) $y = 2x - 25x^2$

KM0336

13. During a projectile motion, if the maximum height equals the horizontal range, then the angle of projection with the horizontal is

(A) $\tan^{-1}(1)$ (B) $\tan^{-1}(2)$ (C) $\tan^{-1}(3)$ (D) $\tan^{-1}(4)$

KM0337

14. A body is thrown horizontally from the top of a tower of height 5 m. It touches the ground at a distance of 10 m from the foot of the tower. The initial velocity of the body is (g = 10 ms^{-2})

(A) 2.5 ms^{-1} (B) 5 ms^{-1} (C) 10 ms^{-1} (D) 20 ms^{-1}

KM0338

15. A particle (A) is dropped from a height and another particle (B) is thrown in horizontal direction with speed of 5 m/sec from the same height. The correct statement is

(A) Both particles will reach at ground simultaneously

(B) Both particles will reach at ground with same speed

(C) Particle (A) will reach at ground first with respect to particle (B)

(D) Particle (B) will reach at ground first with respect to particle (A)

KM0339

16. An aeroplane flying 490 m above ground level at 100 m/s, releases a block. How far on ground will it strike:

(A) 0.1 km (B) 1 km (C) 2 km (D) None

KM0340

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17. Two thin wood screens A and B are separated by 200 m a bullet travelling horizontally at speed of 600 m/s hits the screen A penetrates through it and finally emerges out from B making holes in A and B the resistance of air and wood are negligible the difference of heights of the holes in A and B is.

(A) 5 m (B) $\frac{49}{90}$m (C) $\frac{7}{\sqrt{90}}$m (D) Zero

KM0341

18. A particle is projected from ground with some initial velocity making an angle of 45° with the horizontal. It reaches a height of 7.5 m above the ground while it travels a horizontal distance of 10 m from the point of projection. The initial speed of the projection is:

(A) 5 m/s (B) 10 m/s (C) 20 m/s (D) 40 m/s

KM0342

19. A particle is projected under gravity with velocity $\sqrt{2ag}$ from a point at a height h above the level plane at an angle θ to it. The maximum range R on the ground is:

(A) $\sqrt{(a^2+1)h}$ (B) $\sqrt{a^2h}$ (C) $\sqrt{ah}$ (D) $2\sqrt{a(a+h)}$

KM0343

20. A particle is projected from ground at an angle 45° with initial velocity $20\sqrt{2}$ ms^{-1}. The magnitude of average velocity in a time interval from t = 0 to t = 3 s in ms^{-1} is:

(A) 20.62 (B) 10.31 (C) 41.14 (D) 5.15

KM0344

21. A ball is thrown with a velocity of u making an angle θ with the horizontal. Its velocity vector normal to initial vector (u) after a time interval of:

(A) $\frac{u\sin\theta}{g}$ (B) $\frac{u}{g\cos\theta}$ (C) $\frac{u}{g\sin\theta}$ (D) $\frac{u\cos\theta}{g}$

KM0345

22. A projectile is fired with a velocity u making an angle θ with the horizontal. What is the magnitude of change in velocity when it is at the highest point ?

(A) u cos θ (B) u (C) u sin θ (D) u cos θ – u

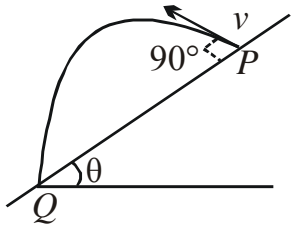
KM0190

23. A ball is projected horizontally. After 3 s from projection its velocity becomes 1.25 times of the velocity of projection. Its velocity of projection is :-

(A) 10 m/s (B) 20 m/s (C) 30 m/s (D) 40 m/s

KM0159

24. In the given figure, if time taken by the projectile to reach Q is T, than PQ =



(A) $Tv\sin\theta$ (B) $Tv\cos\theta$ (C) $Tv\sec\theta$ (D) $Tv\tan\theta$

KM0161

25. A particle is projected up the incline such that its component of velocity along the incline is 10m/s. Time of flight is 2 second and maximum perpendicular distance during the motion from the incline is 5 m. Then velocity of projection will be :-

(A) 10 m/s (B) $10\sqrt{2}$ m/s

(C) $5\sqrt{5}$ m/s (D) none of these **KM0162**

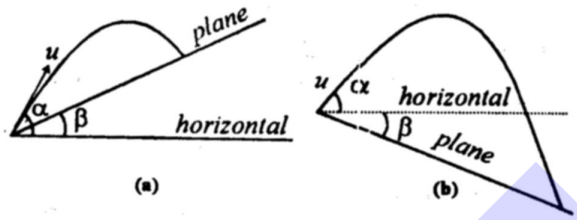
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26. A plane surface is inclined making an angle θ with the horizontal. From the bottom of this inclined plane, a bullet is fired with velocity v. The maximum possible range of the bullet on the inclined plane is

(A) $\frac{v^2}{g}$ (B) $\frac{v^2}{g(1+\sin\theta)}$ (C) $\frac{v^2}{g(1-\sin\theta)}$ (D) $\frac{v^2}{g(1+\cos\theta)}$

KM0346

27. A particle of projected at an angle 'α' to the horizon, up and down is a plane, inclined at an angle β to the horizontal. If the ratio of time flights be 1 : 2, then the ratio $\frac{\tan\alpha}{\tan\beta}$ is equal to



(A) $\frac{2}{1}$ (B) $\frac{3}{1}$ (C) $\frac{4}{1}$ (D) $\frac{5}{3}$

KM0347

28. A particle *A* is projected with speed v_A from a point making an angle 60° with the horizontal. At the same instant, a second particle *B* is thrown vertically upward from a point directly below the maximum height point of parabolic path of *A* with velocity v_B. If the two particles collide then the ratio of v_A/v_B should be :-

(A) 1 (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\sqrt{3}$ **KM0163**

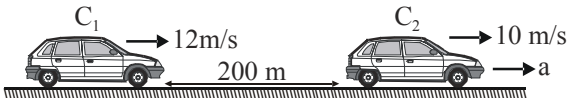
Relative Motion

29. **Statement-1 :** The magnitude of relative velocity of A with respect to B will be always less than V_A.
Statement-2 : The relative velocity of A with respect to B is given by $\vec{V}_{AB} = \vec{V}_A - \vec{V}_B$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

KM0348

30. Two cars C_1 and C_2 are moving in the same direction on a straight single lane road with constant velocities 12 m/s and 10 m/s respectively. When the separation between the two was 200 m, C_2 started accelerating to avoid collision. The minimum acceleration of car C_2 so that they don't collide, is:



(A) 1 cm/s² (B) 2 cm/s² (C) 3 cm/s² (D) 4 cm/s²

KM0349

31. A coin is released inside a lift at a height of 2 m from the floor of the lift. The height of the lift is 10 m. The lift is moving with an acceleration of 9 m/s² downwards. The time after which the coin will strike with the floor of lift is : (g = 10 m/s²)

(A) 4 s (B) 2 s (C) $\frac{4}{\sqrt{21}}$ s (D) $\frac{2}{\sqrt{11}}$ s

KM0350

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32. Two bodies are projected vertically upwards from one point with the same initial velocities v_0, the τ s after the first. How long after will the bodies meet ?

(A) τ (B) $\frac{v_0}{g}+\frac{\tau}{2}$ (C) $\frac{v_0}{g}$ (D) $\frac{\tau}{2}$

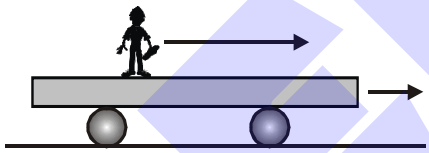
KM0351

33. A car travelling at a constant speed of 20 m/s overtakes another car which is moving at a constant acceleration of 2 m/s^2. The second car was initially at rest. Assume that the length of each car to be 5 m and that at the start of overtaking, the front of the rear car is directly in line with the back of the front car. The total minimum road distance used in overtaking is

(A) 395 m (B) 15.26 m (C) 200.00 m (D) 186.04 m

KM0352

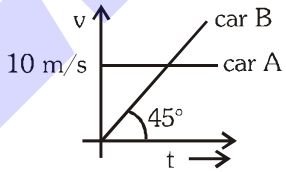
34. A trolley is moving horizontally with a constant velocity of v with respect to earth. A man starts running from one end of the trolley with a velocity 1.5 v with respect to trolley. After reaching the opposite end, the man returns back and continues running with a velocity of 1.5 v w.r.t. the trolley in the backward direction. If the length of the trolley is L then the displacement of the man with respect to earth during the process will be :-



(A) 2.5 L (B) 1.5 L (C) $\frac{5L}{3}$ (D) $\frac{4L}{3}$

KM0165

35. Initially car A is 10.5 m ahead of car B. Both start moving at time t=0 in the same direction along a straight line. The velocity time graph of two cars is shown in figure. The time when the car B will catch the car A, will be :



(A) t = 21 sec (B) t = $2\sqrt{5}$ sec (C) 20 sec (D) None of these

KM0353

36. Some persons hire a boat for 4 hours. The river flows at a speed of 5 km/hr and the boat moves with speed of 15 km/hr relative to the water. How far along the flow direction, they can go if they have to return in 4 hrs.

(A) 5 km (B) 7.5 km (C) $\frac{80}{3}$ km (D) 15 km

KM0354

37. When two bodies approach each other with the different speeds, the distance between them decreases by 120 m for every 1 min. The speeds of the bodies are

(A) 2 m/s and 0.5 m/s (B) 3 m/s and 2 m/s
(C) 1.75 m/s and 0.25 m/s (D) 2.5 m/s and 0.5 m/s

KM0355

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38. Two boats were going downstream with different velocities. When one overtook the other a plastic ball was dropped from one of the boats. Some time later both boats turned back simultaneously & went at the same speeds as before (relative to the water) towards the spot where the ball had been dropped. Which boat will reach the ball first?

(A) The boat which has greater velocity (relative to water)

(B) The boat which has lesser velocity (relative to water)

(C) Both will reach the ball simultaneously

(D Cannot be decided unless we know the actual values of the velocities and the time after which they turned around.

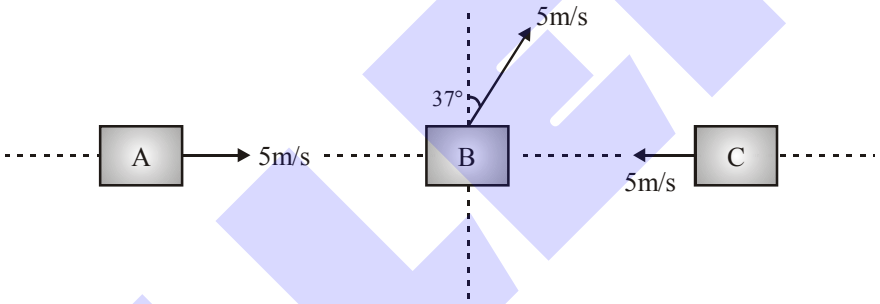
KM0356

39. Men are running in a line along a road with velocity 9 Km/hr behind one another at equal intervals of 20m. Cyclist are also riding along the same line in the same direction at 18 Km/hr at equal intervals of 30m. The speed with which an observer must travel along the road so that whenever he meets a runner he also meet a cyclist is

(A) 9km/h (B) 12 Km/h (C) 18 Km/h (D) 6 Km/h

KM0357

40. Consider the motion of three bodies as shown for an observer on B, what is the magnitude of relative velocity of A with respect to C ?



(A) 10 m/s (B) 0 (C) $\sqrt{20}$ m/s (D) 8 m/s **KM0164**

41. A ship is travelling due east at 10 km/h. A ship heading 30° east of north is always due north from the first ship. The speed of the second ship in km/h is -

(A) $20\sqrt{2}$ (B) $20\sqrt{3/2}$ (C) 20 (D) $20/\sqrt{2}$

KM0358

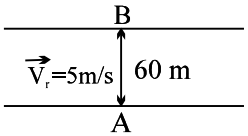
42. A bird is flying towards south with a velocity 40km/h and a train is moving with a velocity 40 km/h towards east. What is the velocity of the bird w.r.t. an obserber in train :

(A) $40\sqrt{2}$ km/h. North – East (B) $40\sqrt{2}$ km/h. South – East

(C) $40\sqrt{2}$ km/h. South – West (D) $40\sqrt{2}$ km/h. North – West

KM0359

43. A man is crossing a river flowing with velocity of 5 m/s. He reaches a point directly across at a distance of 60 m in 5 sec. His velocity in still water should be



(A) 12 m/s (B) 13 m/s (C) 5 m/s (D) 10 m/s

KM0360

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44. A boat of sent across (perpendicular) a river with a velocity 8kph. If the resultant velocity of boat is 10kph, the river is flowing with velocity

(A) 6 kph (B) $\sqrt{13}$ kph (C) $\sqrt{6}$ kph (D) $\sqrt{109}$ kph

KM0361

45. A man wishes to cross a river in a boat. If he crosses the river in minimum time he takes 10 minutes with a drift of 120 m. If he crosses the river taking shortest route, he takes 12.5 minutes, find velocity of the boat with respect to water

(A) 20 m/min (B) 12 m/min (C) 10 m/min (D) 8 m/min

KM0362

46. An aeroplane is flying with the velocity of $V_1 = 800$kmph relative to the air towards south. A wind with velocity of $V_2 = 15ms^{-1}$ is blowing from west to east. What is the velocity of the aeroplane with respect to the earth.

(A) 221.7 ms^{-1} (B) 150 ms^{-1} (C) 82 ms^{-1} (D) 40ms^{-1}

KM0363

47. A motor boat is to reach at a point 30° upstream (w.r.t. normal) on other side of a river flowing with velocity 5m/s. The angle 30° is measured from a direction perpendicular to river flow. Velocity of motorboat with respect to water is $5\sqrt{3}$ m/s. The driver should steer the boat at an angle

(A) 120° with respect to stream direction.
(B) 30° with respect to the perpendicular to the bank.
(C) 30° with respect to the line of destination from starting point.
(D) None of these.

KM0172

48. Wind is blowing in the north direction at speed of 2 m/s which causes the rain to fall at some angle with the vertical. With what velocity should a cyclist drive so that the rain appears vertical to him :

(A) 2 m/s south (B) 2 m/s north (C) 4 m/s west (D) 4 m/s south

KM0169

49. Raindrops are falling vertically with a velocity of 10 m/s. To a cyclist moving on a straight road the raindrops appear to be coming with a velocity of 20 m/s. The velocity of cyclist is :

(A) 10 m/s (B) $10\sqrt{3}$ m/s (C) 20 m/s (D) $20\sqrt{3}$ m/s

KM0364

50. A person is standing on a trolley which moves horizontally with uniform velocity $5\hat{i}$ m/s. At $t = 0$, person throws a ball with velocity $(5\cos\theta\hat{i} + 5\sin\theta\hat{j})$ m/s wrt himself. He always sees ball overhead. Then

(A) $\theta = \frac{\pi}{2}$ (B) $\theta = \frac{\pi}{4}$ (C) $\theta = \pi$ (D) $\theta = \frac{\pi}{3}$

KM0365

51. Two ships are 10 km apart on a line from south to north. The one farther north is moving towards west at 40 kmph and the other is moving towards north at 40 kmph. Then distance of their closest approach is:

(A) 10 km (B) $10\sqrt{2}$ km (C) $\frac{10}{\sqrt{2}}$ km (D) 20 km

KM0366

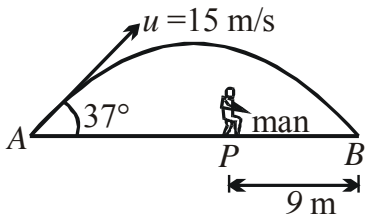
52. Two trucks are moving on parallel tracks. A person on one truck projects a ball vertically upward then path of the ball as seen by four observers: from the ground, from the second truck moving with same velocity as that first truck, from the second truck moving with speed greater than first one in same direction and from the second truck moving with speed less than the first truck in same direction are:

(A) Parabola, Parabola, Parabola and Parabola
(B) Straight line, Straight line, Parabola and Parabola
(C) Parabola, Straight line, Parabola and Parabola
(D) None of these

KM0204

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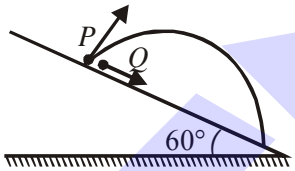
53. A ball is hit by a batsman at an angle of 37° as shown in figure. The man standing at *P* should run at what minimum velocity so that he catches the ball before it strikes the ground. Assume that height of man is negligible in comparison to maximum height of projectile.



(A) 3 m/s (B) 5 m/s (C) 9 m/s (D) 12 m/s

KM0158

54. A particle *P* is projected from a point on the surface of smooth inclined plane (see figure). Simultaneously another particle *Q* is released on the smooth inclined plane from the same position. *P* and *Q* collide after t = 4 s. The speed of projection of *P* is :-



(A) 5 m/s (B) 10 m/s (C) 15 m/s (D) 20 m/s

KM0160

MULTIPLE CORRECT TYPE QUESTIONS

General 2-D Motion

55. A particle moves in the *xy*-plane and at time *t* is at the point (t^2, $t^3 - 2t$). Then

(A) At $t = 2/3$ s, directions of velocity and acceleration are perpendicular

(B) At $t = 0$, directions of velocity and acceleration are perpendicular

(C) At $t = \sqrt{2/3}$ s, particle is moving parallel to *x*-axis

(D) Acceleration of the particle when it is at point (4, 4) is $\left(2\hat{i} + 12\hat{j}\right) m/s$ **KM0174**

56. A point mass is moving in the x-y plane. Its acceleration is a constant vector perpendicular to the x-axis. Which of the following do/does not change with time?

(A) only y-component of its velocity vector

(B) only x-component of its velocity vector

(C) only y-component of its acceleration vector

(D) only x-component of its acceleration vector **KM0213**

Projectile Motion

57. Two particles *A* and B projected along different directions from the same point *P* on the ground with the same velocity of 70 m/s in the same vertical plane . They hit the ground at the same point *Q* such that *PQ* = 480 m. Then [g = 9.8 m/s^2]

(A) Ratio of their times of flight is 4 : 5

(B) Ratio of their maximum heights is 9 : 16

(C) Ratio of their minimum speeds during flights is 4 : 3

(D) The bisector of the angle between their directions of projection makes 45° with horizontal

KM0176

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58. If T is the total time of flight, H the maximum height and R is the horizontal range of a projectile. Then x and y co–ordinates at any time t are related as :

(A) $y = 4H\left(\frac{t}{T}\right)\left(1-\frac{t}{T}\right)$ (B) $y = 4H\left(\frac{T}{t}\right)\left(1-\frac{T}{t}\right)$

(C) $y = 4H\left(\frac{x}{R}\right)\left(1-\frac{x}{R}\right)$ (D) $y = 4H\left(\frac{R}{x}\right)\left(1-\frac{R}{x}\right)$

KM0367

59. A ball is thrown from ground such that it just crosses two poles of equal height kept 80 m apart. The maximum height attained by the ball is 80 m. When the ball passes the first pole, its velocity makes 45° with horizontal. The correct alternatives is/are :- ($g = 10\ m/s^2$)

(A) Time interval between the two poles is 4 s.

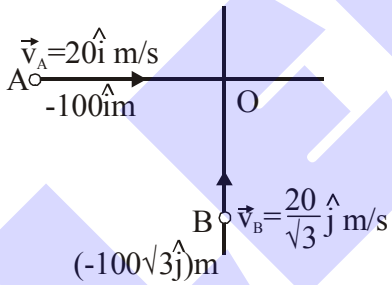
(B) Height of the pole is 60 m.

(C) Range of the ball is 160 m.

(D) Angle of projection is $\tan^{-1}(2)$ with horizontal. **KM0214**

Relative Motion

60. Positions of two vehicles A and B with reference to origin O and their velocities are as shown.



(A) they will collide (B) distance of closest approach is 100 m.

(C) their relative speed is $\frac{40}{\sqrt{3}}$ m/s (D) their relative velocity is $\frac{20}{\sqrt{3}}$ m/s

KM0177

61. A man on a rectilinearly moving cart, facing the direction of motion, throws a ball straight up with respect to himself

(A) The ball will always return to him.

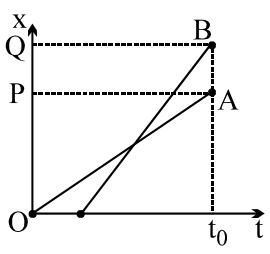
(B) The ball will never return to him.

(C) The ball will return to him if the cart moves with constant velocity.

(D) The ball will fall behind him if the cart moves with some positive acceleration.

KM0218

62. The position–time (x–t) graphs for two children A and B returning from their school O to their homes P and Q respectively along straight line path (taken as x axis) are shown in figure below. Choose the **CORRECT** statement (s):



(A) A lives closer to the school than B

(B) A starts from the school earlier than B

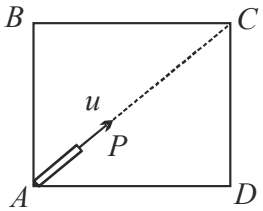
(C) A and B have equal average velocities from 0 to t_0.

(D) B overtakes A on the way

KM0095

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63. A large rectangular box moves vertically downward with an acceleration a. A toy gun fixed at *A* and aimed towards *C* fires a particle *P*.



(A) *P* will hit *C* if $a = g$

(B) *P* will hit the roof *BC*, if $a > g$

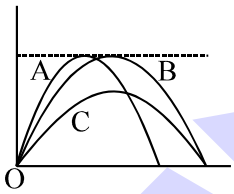
(C) *P* will hit the wall *CD* if $a < g$

(D) May be either (A), (B) or (C), depending on the speed of projection of *P*

KM0220

MATRIX MATCH TYPE QUESTION

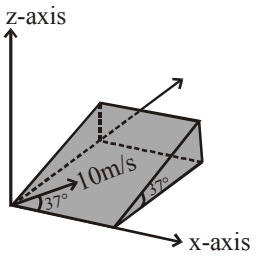
64. Trajectories are shown in figure for three kicked footballs. Initial vertical & horizontal velocity components are u_y and u_x respectively. Ignoring air resistance, choose the correct statement from **Column-II** for the value of variable in **Column-I**.



	Column-I		**Column-II**
(A)	time of flight	(P)	greatest for A only
(B)	u_y/u_x	(Q)	greatest for C only
(C)	u_x	(R)	equal for A and B
(D)	$u_x u_y$	(S)	equal for B and C

KM0182

65. A small ball is projected along the surface of a smooth inclined plane with speed 10m/s along the direction shown at t = 0. The point of projection is origin, z-axis is along vertical. The acceleration due to gravity is 10 m/s^2. Column-I lists values of certain parameters related to motion of ball and column-II lists different time instants. Match appropriately.



	Column-I		**Column-II**
(A)	Distance from x-axis is 2.25m	(P)	0.5 s
(B)	Speed is minimum	(Q)	1.0 s
(C)	Velocity makes angle 37° with x-axis	(R)	1.5 s
		(S)	2.0 s

KM0224

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EXERCISE (S)

General 2-D motion

1. The position of a particle is given by

$$r = 3.0t\,\hat{i} - 2.0t^2\,\hat{j} + 4.0\ k\ m$$

where t is in seconds and the coefficients have the proper units for r to be in metres.
(a) Find the v and a of the particle?
(b) What is the magnitude and direction of velocity of the particle at t = 2.0 s ? **KM0109**

2. The vertical height y and horizontal distance x of a projectile on a certain planet are given by x = (3t)m, y = $(4t - 6t^2)$ m where t is in seconds. Find the speed of projection (in m/s).

KM0108

Projectile motion

3. A particle is projected with a speed of 10 m/s at an angle 37° with the vertical. Find (i) time of flight (ii) maximum height above ground (iii) horizontal range. **KM0112**

4. A cricketer can throw a ball to a maximum horizontal distance of 100 m. How much high above the ground can the cricketer throw the same ball ? **KM0114**

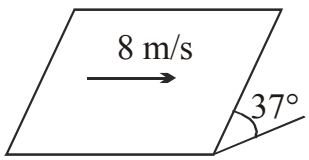
5. A particle is projected in *x*-*y* plane with *y*-axis along vertical, the point of projection is origin. The equation of a path is $y = \sqrt{3}\,x - \frac{gx^2}{2}$. Find angle of projection and speed of projection.

KM0116

6. A Bomber flying upward at an angle of 53° with the vertical releases a bomb at an altitude of 800 m. The bomb strikes the ground 20 s after its release. Find: [Given sin 53° = 0.8; g = 10 m/s^2]
(i) The speed of the bomber at the time of release of the bomb .
(ii) The maximum height attained by the bomb .
(iii) The horizontal distance travelled by the bomb before it strikes the ground
(iv) The velocity (magnitude & direction) of the bomb just when it strikes the ground . **KM0120**

7. A person standing on the top of a cliff 171 ft high has to throw a packet to his friend standing on the ground 228 ft horizontally away. If he throws the packet directly aiming at the friend with a speed of 15.0 ft/s, how short will the packet fall ? **KM0368**

8. A ball is projected on smooth inclined plane in direction perpendicular to line of greatest slope with velocity of 8 m/s. Find it's speed after 1 s.



KM0125

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Relative Motion

9. A stone is dropped from the top of a tall cliff, and 1s later a second stone is thrown vertically downward with a velocity of 20 ms^{-1}. How far below the top of the cliff will the second stone overtake the first?

KM0010

10. A car travelling at 60 km/hr overtakes another car travelling at 42 km/h. Assuming each car to be 5.0 m long, find the time taken during the overtake and the total road distance used for the overtake.

KM0369

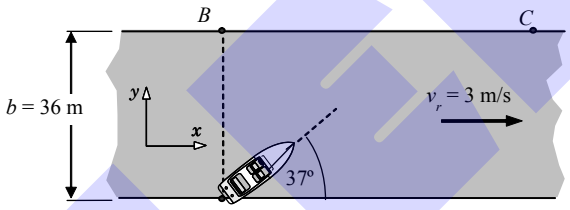
11. Two body move from the same point along a straight line. The first body moves with velocity $v = (3t^2 - 6t)$ m/s, the second with velocity $v = (10t + 20)$ m/s. At what instant and at what distance from the initial point will they meet.

KM0044

12. A man crosses a river by a boat. If he crosses the river in minimum time he takes 10 minutes with a drift 120 m. If he crosses the river taking shortest path, he takes 12.5 minutes. Assuming $v_{b/r} > v_r$, find
(i) width of the river,
(ii) velocity of the boat with respect to water ($v_{b/r}$)
(iii) speed of the current (v_r)

KM0131

13. Velocity of the boat with respect to river is 10 m/s. From point A it is steered in the direction shown to reach point C. Find the time of the trip and distance between B and C.



KM0135

14. An aeroplane has to go from a point A to another point B, 500 km away due 30° east of north. A wind is blowing due north at a speed of 20 m/s. The air-speed of the plane is 150 m/s.
(a) Find the direction in which the pilot should head the plane to reach the point B.
(b) Find the time taken by the plane to go from A to B.

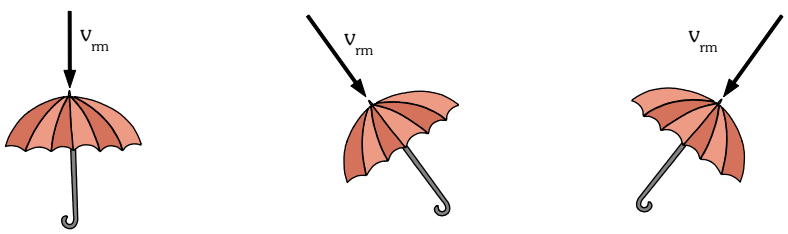
KM0370

Paragraph for Question no. 15 to 18: Rain and man

By the term velocity of rain, we mean velocity with which raindrops fall relative to the ground. In absence of wind, raindrops fall vertically and in presence of wind raindrops fall obliquely. Moreover raindrops acquire a constant terminal velocity due air resistance very quickly as they fall toward the earth. A moving man relative to himself observes an altered velocity of raindrops, which is known as velocity of rain relative to the man. It is given by the following equation.

$$\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$$

A standstill man relative to himself observes rain falling with velocity, which is equal to velocity of the raindrops relative to the ground. To protect himself a man should hold his umbrella against velocity of raindrops relative to himself as shown in the following figure.



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15. Rain is falling vertically with velocity 80 cm/s.
(a) How should you hold your umbrella? (in standing position)
(b) You start walking towards the east with velocity 60 cm/s. How should you hold your umbrella?
(c) You are walking towards the west with velocity 60 cm/s. How should you hold your umbrella?
(d) You are walking towards the north with velocity 60 cm/s. How should you hold your umbrella?
(e) You are walking towards the south with velocity 80 cm/s. How should you hold your umbrella?
KM0180

16. When you are standstill in rain, you have to hold your umbrella vertically to protect yourself.
(a) When you walk with velocity 90 cm/s, you have to hold your umbrella at 53° above the horizontal. What is velocity of the raindrops relative to the ground and relative to you?
(b) If you walk with speed 160 cm/s, how should you hold your umbrella?
KM0180

17. A man walks in rain at 72 cm/s due east and observes the rain falling vertically. When he stops, rain appears to strike his back at 37° from the vertical. Find velocity of raindrops relative to the ground.
KM0180

18. When you walk in rain at 75 cm/s, you have to hold your umbrella vertically and when you double your speed in the same direction, you have to hold your umbrella at 53° above the horizontal. What is the rain velocity?
KM0180

19. Rain is falling vertically with a speed of 20 m/s relative to air. A person is running in the rain with a velocity of 5 m/s and a wind is also blowing with a speed of 15 m/s (both towards east). Find the angle with the vertical at which the person should hold his umbrella for best protection from rain.
KM0132

Paragraph for Question no. 20 to 22 : Flag in wind

When you are standstill holding a flag, the flag flutters in the direction of wind. When you start running the direction of fluttering of the flag changes in to the direction of the wind relative to you. In all case a flag flutters in the direction of the wind relative to the flag.

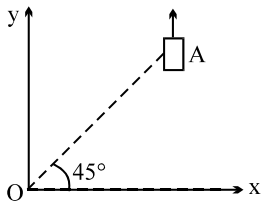
20. When you are standstill holding a flag the flag flutters in the north and when you run at 8 m/s due east, the flag flutters in direction 37° north of west. Find the wind velocity. **KM0181**

21. Wind is blowing uniformly due north everywhere with velocity 12 m/s. A car mounted with a flag starts running towards east. After 9 s from start the flag flutters in 53° north of west and after 16 s from the start the flag flutters in 37° north of west.
(a) Find velocity of the car 9 s after it starts.
(b) Find velocity of the car 16 s after it starts.
(c) If the car maintains uniform acceleration, find acceleration of the car. **KM0181**

22. Holding a flag, when you run at 8 m/s due east, the flag flutters in the north and when you run at 2 m/s due south, the flag flutters in the northeast. If the wind velocity is uniform and remain constant, find the wind velocity. **KM0181**

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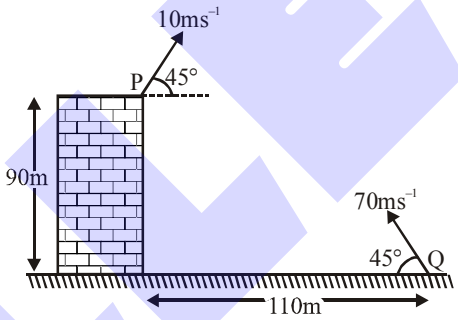
23. On a frictionless horizontal surface, assumed to be the x-y plane, a small trolley. A is moving along a straight line parallel to the y-axis (see figure) with a constant velocity of $(\sqrt{3}-1)$ m/s. At a particular instant, when the line OA makes an angle of 45° with the x-axis, a ball is thrown along the surface from the origin O. Its velocity makes an angle ϕ with the x-axis and it hits the trolley.



(a) The motion of the ball is observed from the frame of trolley. Calculate the angle θ made by the velocity vector of the ball with the x-axis in this frame.

(b) Find the speed of the ball with respect to the surface, if $\phi = \frac{4\theta}{3}$. **KM0128**

24. Two particles P and Q are launched simultaneously as shown in figure. Find the minimum distance between particles in meters.



KM0130

25. A train is moving along a straight line with a constant acceleration 'a'. A boy standing in the train throws a ball forward with a speed of 10 m/s, at an angle of 60° to the horizontal. The boy has to move forward by 1.15 m inside the train to catch the ball back at the initial height. The acceleration of the train, in m/s^2, is **[IIT-JEE 2011]**

KM0232

26. Two guns, situated at the top of a hill of height 10 m, fire one shot each with the same speed $5\sqrt{3}$ m/s at some interval of time. One gun fires horizontally and other fires at an angle of 60° up the horizontal. The shots collide in air at a point *P*. Find
(i) the time interval between the firings, and (ii) the coordinates of the point *P*.
(Take origin of the coordinates system at the foot of the hill vertically below the muzzle and trajectories in *X*-*Y* plane) **KM0144**

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EXERCISE (JM)

1. A projectile is given an initial velocity of $(\hat{i}+2\hat{j})$ m/s, where $\hat{i}$ is along the ground and $\hat{j}$ is along the vertical. If g = 10 m/s^2, the equation of its trajectory is : **[AIEEE - 2013]**

(1) $y = x - 5x^2$ (2) $y = 2x - 5x^2$ (3) $4y = 2x - 5x^2$ (4) $4y = 2x - 25x^2$

KM0230

SELECTED PROBLEMS FROM JEE-MAINS ONLINE PAPERS

2. The position co-ordinates of a particle moving in a 3-D coordinate system is given by

$x = a\cos\omega t$

$y = a\sin\omega t$

and $z = a\omega t$

The speed of the particle is : **[JEE-Main-2019_Jan]**

(1) $a\omega$ (2) $\sqrt{3}\,a\omega$ (3) $\sqrt{2}\,a\omega$ (4) $2a\omega$

KM0371

3. Two guns A and B can fire bullets at speeds 1 km/s and 2 km/s respectively. From a point on a horizontal ground, they are fired in all possible directions. The ratio of maximum areas covered by the bullets fired by the two guns, on the ground is : **[JEE-Main-2019_Jan]**

(1) 1 : 2 (2) 1 : 4 (3) 1 : 8 (4) 1 : 16

KM0372

4. A passenger train of length 60m travels at a speed of 80 km/hr. Another freight train of length 120 m travels at a speed of 30 km/hr. The ratio of times taken by the passenger train to completely cross the freight train when : (i) they are moving in the same direction, and (ii) in the opposite directions is :

[JEE-Main-2019_Jan]

(1) $\frac{5}{2}$ (2) $\frac{25}{11}$ (3) $\frac{3}{2}$ (4) $\frac{11}{5}$

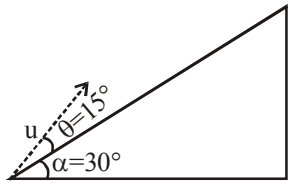
KM0373

5. Ship A is sailing towards north-east with velocity $\vec{v} = 30\hat{i} + 50\hat{j}$ km/hr where $\hat{i}$ points east and $\hat{j}$, north. Ship B is at a distance of 80 km east and 150 km north of Ship A and is sailing towards west at 10 km/hr. A will be at minimum distance from B in : **[JEE-Main-2019_April]**

(1) 4.2 hrs. (2) 2.2 hrs. (3) 3.2 hrs. (4) 2.6 hrs.

KM0374

6. A plane is inclined at an angle $\alpha = 30°$ with a respect to the horizontal. A particle is projected with a speed u = 2 ms^{-1} from the base of the plane, making an angle $\theta = 15°$ with respect to the plane as shown in the figure. The distance from the base, at which the particle hits the plane is close to : (Take g = 10 ms^{-2}) **[JEE-Main-2019_April]**



(1) 14 cm (2) 20 cm (3) 18 cm (4) 26 cm

KM0375

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7. A particle is moving along the x-axis with its coordinate with the time 't' given be $x(t) = 10 + 8t - 3t^2$. Another particle is moving the y-axis with its coordinate as a function of time given by $y(t) = 5 - 8t^3$. At t = 1s, the speed of the second particle as measured in the frame of the first particle is given as $\sqrt{v}$. Then v (in m/s) is _____. **[JEE-Main-2020_Jan]**

KM0376

8. Starting from the origin at time t = 0, with initial velocity $5\hat{j}\ ms^{-1}$, a particle moves in the x-y plane with a constant acceleration of $(10\hat{i}+4\hat{j})ms^{-2}$. At time t, its coordinates are (20 m, y_0 m). The values of t and y_0, are respectively :

[JEE-Main-2020_Sep]

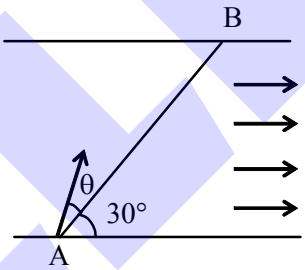
KM0377

(1) 4s and 52 m (2) 2s and 24 m (3) 2s and 18 m (4) 5s and 25 m

9. A swimmer can swim with velocity of 12 km/h in still water. Water flowing in a river has velocity 6 km/h. The direction with respect to the direction of flow of river water he should swim in order to reach the point on the other bank just opposite to his starting point is _____°. (Round off to the Nearest Integer) (find the angle in degree) **[JEE-Main-2021_March]**

KM0378

10. A swimmer wants to cross a river from point A to point B. Line AB makes an angle of 30° with the flow of river. Magnitude of velocity of the swimmer is same as that of the river. The angle θ with the line AB should be ____°, so that the swimmer reaches point B. **[JEE-Main-2021_July]**



KM0379

11. A bomb is dropped by fighter plane flying horizontally. To an observer sitting in the plane, the trajectory of the bomb is a :

(1) hyperbola

(2) parabola in the direction of motion of plane

(3) straight line vertically down the plane

(4) parabola in a direction opposite to the motion of plane **[JEE-Main-2021_Aug]**

KM0380

12. A boy reaches the airport and finds that the escalator is not working. He walks up the stationary escalator in time t_1. If he remains stationary on a moving escalator then the escalator takes him up in time t_2. The time taken by him to walk up on the moving escalator will be :

[JEE-Main-2021_July]

(1) $\frac{t_1 t_2}{t_2 - t_1}$ (2) $\frac{t_1 + t_2}{2}$ (3) $\frac{t_1 t_2}{t_2 + t_1}$ (4) $t_2 - t_1$

KM0381

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13. In a car race on straight road, car A takes a time t less than car B at the finish and passes finishing point with a speed 'v' more than that of car B. Both the cars start from rest and travel with constant acceleration a_1 and a_2 respectively. Then 'v' is equal to : **[JEE-Main-2019_Jan]**

(1) $\frac{a_1 + a_2}{2}t$ (2) $\sqrt{2a_1a_2}t$ (3) $\frac{2a_1a_2}{a_1 + a_2}t$ (4) $\sqrt{a_1a_2}t$

KM0382

14. A projectile is projected with velocity of 25 m/s at an angle θ with the horizontal. After t seconds its inclination with horizontal becomes zero. If R represents horizontal range of the projectile, the value of θ will be : [use g = 10 m/s^2] **[JEE-Main-2022_June]**

(1) $\frac{1}{2}\sin^{-1}\left(\frac{5t^2}{4R}\right)$ (2) $\frac{1}{2}\sin^{-1}\left(\frac{4R}{5t^2}\right)$ (3) $\tan^{-1}\left(\frac{4t^2}{5R}\right)$ (4) $\cot^{-1}\left(\frac{R}{20t^2}\right)$

KM0383

15. A girl standing on road holds her umbrella at 45° with the vertical to keep the rain away. If she starts running without umbrella with a speed of $15\sqrt{2}\,\text{kmh}^{-1}$, the rain drops hit her head vertically. The speed of rain drops with respect to the moving girl is : **[JEE-Main-2022_June]**

(1) 30 kmh^{-1} (2) $\frac{25}{\sqrt{2}}kmh^{-1}$ (3) $\frac{30}{\sqrt{2}}kmh^{-1}$ (4) 20 kmh^{-1}

KM0384

16. A person can throw a ball upto a maximum range of 100 m. How high above the ground he can throw the same ball? **[JEE-Main-2022_June]**

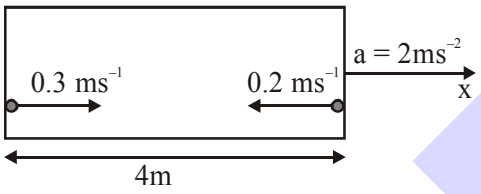
(1) 25 m (2) 50 m (3) 100 m (4) 200 m

KM0385

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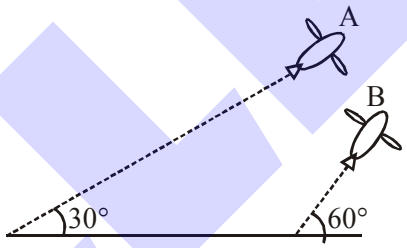
EXERCISE (JA)

1. A rocket is moving in a gravity free space with a constant acceleration of 2 ms^{-2} along + x direction (see figure). The length of a chamber inside the rocket is 4m. A ball is thrown from the left end of the chamber in + x direction with a speed of 0.3 ms^{-1} relative to the rocket. At the same time, another ball is thrown in –x direction with a speed of 0.2 ms^{-1} from its right end relative to the rocket. The time in seconds when the two balls hit each other is **[JEE Advanced 2014]**



KM0233

2. Airplanes A and B are flying with constant velocity in the same vertical plane at angles 30° and 60° with respect to the horizontal respectively as shown in figure. The speed of A is $100\sqrt{3}$ ms^{-1}. At time t = 0 s, an observer in A finds B at a distance of 500 m. This observer sees B moving with a constant velocity perpendicular to the line of motion of A. If at $t = t_0$, A just escapes being hit by B, t_0 in seconds is **[JEE Advanced-2014]**



KM0234

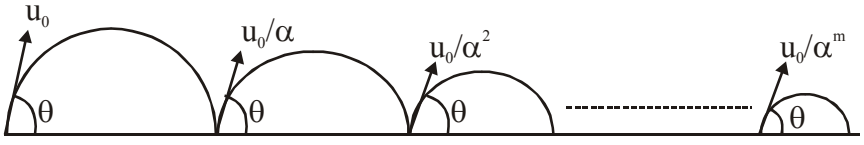
3. A ball is projected from the ground at an angle of 45° with the horizontal surface. It reaches a maximum height of 120 m and returns to the ground. Upon hitting the ground for the first time, it loses half of its kinetic energy. Immediately after the bounce, the velocity of the ball makes an angle of 30° with the horizontal surface. The maximum height it reaches after the bounce, in metres, is.....

[JEE Advanced-2018]

KM0235

4. A ball is thrown from ground at an angle θ with horizontal and with an initial speed u_0. For the resulting projectile motion, the magnitude of average velocity of the ball up to the point when it hits the ground for the first time is V_1. After hitting the ground, ball rebounds at the same angle θ but with a reduced speed of u_0/α. Its motion continues for a long time as shown in figure. If the magnitude of average velocity of the ball for entire duration of motion is 0.8 V_1, the value of α is_____

[JEE Advanced-2019]



KM0386

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5. Starting at time t = 0 from the origin with speed 1 ms^{-1}, a particle follows a two-dimensional trajectory in the x-y plane so that its coordinates are related by the equation $y = \frac{x^2}{2}$. The x and y components of its acceleration are denoted by a_x and a_y, respectively. Then **[JEE Advanced-2020]**

(A) $a_x = 1\ ms^{-2}$ implies that when the particle is at the origin, $a_y = 1\ ms^{-2}$

(B) $a_x = 0$ implies $a_y = 1\ ms^{-2}$ at all times

(C) at t = 0, the particle's velocity points in the x-direction

(D) $a_x = 0$ implies that at t = 1 s, the angle between the particle's velocity and the x axis is 45°

KM0387

6. A projectile is fired from horizontal ground with speed v and projection angle θ. When the acceleration due to gravity is g, the range of the projectile is d. If at the highest point in its trajectory, the projectile enters a different region where the effective acceleration due to gravity is $g' = \frac{g}{0.81}$, then the new range is d' = nd. The value of n is_____. **[JEE Advanced-2022]**

KM0388

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ANSWER KEY

EXERCISE # O

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	A	A	C	C	C	C	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	D	C	A	B	B	C	D	A
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	C	D	D	B	B	B	B	D	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	B	B	B	D	A	C	C	C	A	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	C	B	A	A	A	C	B	B	A
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	C	C	B	B	A,B,C,D	B,C,D	B,C,D	A,C	A,B,C,D	B,C
Que.	61	62	63	64	65					
Ans.	C,D	A,B,D	A,B	A-R; B-P; C-Q; D-S	A-P,R; B-Q; C--S					

EXERCISE # S

1. (a) $v(t) = \left(3.0\hat{i} - 4.0t\,\hat{j}\right)$ â $(t) = -4.0\,\hat{j}$; (b) 8.54 ms^{-1}, 70° with x-axis **2.** 5

3. (i) 1.6 sec; (ii) 3.2 m; (iii) 9.6 m **4.** 50 m **5.** 60°, 2 m/s.

6. (i) 100 m/s; (ii) 980 m; (iii) 1600 m; (iv) $\left(80\hat{i} - 140\,\hat{j}\right)$ **7.** 192 ft **8.** 10 m/s **9.** $\frac{45}{4}$ m

10. 2 s , 38m **11.** 10 sec ; 700m **12.** 200 m, 20 m/min, 12 m/min **13.** 6 s, 66 m

14. (a) sin^{-1} (1/15) east of the line AB; (b) 50 min

15. (a) Vertically; (b) 53° above east; (c) 53° above west; (d) 53° above north; (e) 45° above south

16. (a) 120 cm/s vertically, 150 cm/s 53° above horizontal; (b) 37° above the horizontal.

17. 120 cm/s **18.** 125 cm/s at 37° from the vertical **19.** tan^{-1}(1/2) **20.** 6 m/s due north

21. (a) 9 m/s; (b) 16 m/s; (c) 1 m/s^2 **22.** 10 m/s, 37° north of east **23.** (a) 45°, (b) 2 m/sec

24. 6 **25.** 5 **26.** (i) 1 s; (ii) (5√3 m, 5 m)

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EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	4	4	4	2	580	3	120	30
Que.	11	12	13	14	15	16				
Ans.	3	3	4	4	3	2				

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6
Ans.	8 or 2	5	29.60 to 30.40	4	A,B,C,D	0.95

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